

14.3 Solution: From the relation

$$\frac{dt}{dt'} = 1 - \boldsymbol{\beta}(t') \cdot \mathbf{n}(t'),$$

we can formally write

$$\frac{d}{dt} = \frac{dt'}{dt} \frac{d}{dt'} = \frac{1}{1 - \boldsymbol{\beta} \cdot \mathbf{n}} \frac{d}{dt'}.$$

From now on, we will drop the explicit t' -dependence in $\boldsymbol{\beta}$ and $\mathbf{n}$.

Let us establish some useful results. First,

$$\frac{d\mathbf{n}}{dt'} = \frac{d}{dt'} \left(\frac{\mathbf{R}}{R} \right) = -\frac{\mathbf{v}}{R} + \frac{\mathbf{R}(\mathbf{R} \cdot \mathbf{v})}{R^3} = \frac{\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v}}{R}, \quad (1)$$

since

$$\frac{dR}{dt'} = -\frac{\mathbf{R} \cdot \mathbf{v}}{R}.$$

Then,

$$\begin{aligned} \frac{d^2\mathbf{n}}{dt'^2} &= \frac{d}{dt'} \left(\frac{\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v}}{R} \right) \\ &= \frac{1}{R} \left[\dot{\mathbf{n}}(\mathbf{n} \cdot \mathbf{v}) + \mathbf{n}(\dot{\mathbf{n}} \cdot \mathbf{v}) + \mathbf{n}(\mathbf{n} \cdot \dot{\mathbf{v}}) - \dot{\mathbf{v}} \right] + \left[\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v} \right] \frac{\mathbf{n} \cdot \mathbf{v}}{R^2}. \end{aligned}$$

Applying Eq. (1), we have

$$\begin{aligned} \dot{\mathbf{n}}(\mathbf{n} \cdot \mathbf{v}) + \mathbf{n}(\dot{\mathbf{n}} \cdot \mathbf{v}) &= \frac{1}{R} \left[(\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v})(\mathbf{n} \cdot \mathbf{v}) + \mathbf{n}(\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v}) \cdot \mathbf{v} \right] \\ &= \frac{1}{R} \left[\mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - \mathbf{v}(\mathbf{n} \cdot \mathbf{v}) + \mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - v^2\mathbf{n} \right] \\ &= \frac{1}{R} \left[2\mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - \mathbf{v}(\mathbf{n} \cdot \mathbf{v}) - v^2\mathbf{n} \right]. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{d^2\mathbf{n}}{dt'^2} &= \frac{\mathbf{n} \times (\mathbf{n} \times \dot{\mathbf{v}})}{R} + \frac{1}{R^2} \left[2\mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - \mathbf{v}(\mathbf{n} \cdot \mathbf{v}) - v^2\mathbf{n} + \mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - \mathbf{v}(\mathbf{n} \cdot \mathbf{v}) \right] \\ &= \frac{\mathbf{n} \times (\mathbf{n} \times \dot{\mathbf{v}})}{R} + \frac{1}{R^2} \left[3\mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - 2\mathbf{v}(\mathbf{n} \cdot \mathbf{v}) - v^2\mathbf{n} \right]. \end{aligned}$$

Now,

$$\begin{aligned} \frac{1}{c^2} \frac{d^2}{dt'^2} [\mathbf{n}]_{\text{ret}} &= \frac{1}{c^2} \frac{1}{1 - \boldsymbol{\beta} \cdot \mathbf{n}} \frac{d}{dt'} \left[\frac{1}{1 - \boldsymbol{\beta} \cdot \mathbf{n}} \frac{d\mathbf{n}}{dt'} \right] \\ &= \frac{1}{c^2} \left[\frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3} (\dot{\boldsymbol{\beta}} \cdot \mathbf{n} + \boldsymbol{\beta} \cdot \dot{\mathbf{n}}) \frac{d\mathbf{n}}{dt'} + \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^2} \frac{d^2\mathbf{n}}{dt'^2} \right] \\ &= \frac{1}{c^2} \left[\frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3} \left(\dot{\boldsymbol{\beta}} \cdot \mathbf{n} + \frac{\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v}}{R} \cdot \boldsymbol{\beta} \right) \frac{\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v}}{R} \right. \\ &\quad \left. + \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^2} \left(\frac{\mathbf{n} \times (\mathbf{n} \times \dot{\mathbf{v}})}{R} + \frac{1}{R^2} [3\mathbf{n}(\mathbf{n} \cdot \mathbf{v})^2 - 2\mathbf{v}(\mathbf{n} \cdot \mathbf{v}) - v^2\mathbf{n}] \right) \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{cR} \left(\frac{\mathbf{n} \cdot \dot{\boldsymbol{\beta}}}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3} [\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}] + \frac{\mathbf{n} \times (\mathbf{n} \times \dot{\boldsymbol{\beta}})}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^2} \right) \\
&\quad + \frac{1}{R^2} \left(\frac{[(\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}) \cdot \boldsymbol{\beta}](\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta})}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3} + \frac{3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - 2\mathbf{v}(\mathbf{n} \cdot \boldsymbol{\beta}) - \beta^2 \mathbf{n}}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^2} \right) \\
&= I_1 + I_2,
\end{aligned}$$

and by direct evaluation,

$$\begin{aligned}
I_1 &= \frac{1}{c(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R} \left[\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})(\mathbf{n} \cdot \dot{\boldsymbol{\beta}}) - \boldsymbol{\beta}(\mathbf{n} \cdot \dot{\boldsymbol{\beta}}) + \mathbf{n}(\mathbf{n} \cdot \dot{\boldsymbol{\beta}}) - \dot{\boldsymbol{\beta}} - \mathbf{n}(\mathbf{n} \cdot \dot{\boldsymbol{\beta}})(\mathbf{n} \cdot \boldsymbol{\beta}) + \dot{\boldsymbol{\beta}}(\mathbf{n} \cdot \boldsymbol{\beta}) \right] \\
&= \frac{1}{c(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R} \left[\mathbf{n}(\mathbf{n} \cdot \dot{\boldsymbol{\beta}}) - \dot{\boldsymbol{\beta}} - \boldsymbol{\beta}(\mathbf{n} \cdot \dot{\boldsymbol{\beta}}) + \dot{\boldsymbol{\beta}}(\mathbf{n} \cdot \boldsymbol{\beta}) \right] \\
&= \frac{1}{c(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R} \left[\mathbf{n} \times (\mathbf{n} \times \dot{\boldsymbol{\beta}}) - \mathbf{n} \times (\boldsymbol{\beta} \times \dot{\boldsymbol{\beta}}) \right] \\
&= \frac{\mathbf{n} \times ((\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}})}{c(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R},
\end{aligned}$$

and

$$\begin{aligned}
I_2 &= \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} [((\mathbf{n} \cdot \boldsymbol{\beta})^2 - \beta^2)(\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}) + (3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta}) - \beta^2 \mathbf{n})(1 - \mathbf{n} \cdot \boldsymbol{\beta})] \\
&= \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} \left[\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 - \beta^2 \mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta})^2 + \beta^2 \boldsymbol{\beta} + 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta}) - \beta^2 \mathbf{n} \right. \\
&\quad \left. - 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 + 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta})^2 + \beta^2 \mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) \right] \\
&= \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} \left[-2\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 - \boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta})^2 + \beta^2 \boldsymbol{\beta} + 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta}) - \beta^2 \mathbf{n} \right].
\end{aligned}$$

We can also calculate the second term of the Feynman expression as

$$\begin{aligned}
\left[\frac{R}{c} \right]_{\text{ret}} \frac{d}{dt} \left[\frac{\mathbf{n}}{R^2} \right]_{\text{ret}} &= \frac{1}{1 - \boldsymbol{\beta} \cdot \mathbf{n}} \frac{R}{c} \left(\frac{1}{R} \frac{d\mathbf{n}}{dt'} + \mathbf{n} \frac{d}{dt'} \left(\frac{1}{R} \right) \right) \\
&= \frac{1}{1 - \boldsymbol{\beta} \cdot \mathbf{n}} \frac{1}{c} \left(\frac{\mathbf{n}(\mathbf{n} \cdot \mathbf{v}) - \mathbf{v}}{R^2} + \frac{2\mathbf{n}(\mathbf{n} \cdot \mathbf{v})}{R^2} \right) \\
&= \frac{3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})R^2},
\end{aligned}$$

evaluated at the retarded time. Include the first term, which is also proportional to $1/R^2$, we will get

$$\begin{aligned}
&\frac{\mathbf{n}}{R^2} + \frac{3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})R^2} + I_2 \\
&= \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} \left[\mathbf{n}(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 + (3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta})(1 - \boldsymbol{\beta} \cdot \mathbf{n})^2 \right. \\
&\quad \left. - 2\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 - \boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta})^2 + \beta^2 \boldsymbol{\beta} + 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta}) - \beta^2 \mathbf{n} \right] \\
&= \frac{1}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} \left[\mathbf{n} - 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) + 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - \mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 \right]
\end{aligned}$$

$$\begin{aligned}
& +3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - 6\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 + 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 - \boldsymbol{\beta} + 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta})^2 \\
& - 2\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^3 - \boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta})^2 + \beta^2\boldsymbol{\beta} + 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})^2 - 2\boldsymbol{\beta}(\mathbf{n} \cdot \boldsymbol{\beta}) - \beta^2\mathbf{n} \Big] \\
= & \frac{(\mathbf{n} - \boldsymbol{\beta})(1 - \beta^2)}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} \\
= & \frac{\mathbf{n} - \boldsymbol{\beta}}{\gamma^2(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2}.
\end{aligned}$$

Combine with I_1 , we will obtain Eq. (14.14),

$$\mathbf{E}(\mathbf{x}, t) = e \left[\frac{\mathbf{n} - \boldsymbol{\beta}}{\gamma^2(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R^2} \right]_{\text{ret}} + \frac{e}{c} \left[\frac{\mathbf{n} \times \left((\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}} \right)}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})^3 R} \right]_{\text{ret}}.$$